



A World You Did Not Author

Modeling Is Cheap, Intent Is Expensive: What a Solved Benchmark Says
to the Anthropology of Agents

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ABSTRACT

AGENTWORLD asks what happens when minds that did not grow up in a world are dropped into it in large numbers. We report on a controlled miniature of exactly that situation. ARC-AGI-3 is an interactive benchmark that places an agent in an unfamiliar grid game with no instructions, no language, and no stated goal; the agent must learn the rules by acting and then reach an end it was never told. Humans clear every game. Frontier agents scored under one percent at launch. Our agent, built on the OPENWORLD framework, completes all 25/25 public games under an audited *source-free* protocol, never reading a game's code: it acts, writes down what it learns as a small verified program, plans inside that program, and replays the plan against the real engine.

The lesson we carry to AGENTWORLD is not the score. It is *where the difficulty sat*. Learning the *dynamics* of a world that did not author you turned out to be cheap: an agent that writes its model as code and checks it against held-out transitions recovers the rules exactly. Learning the *intent* of that world was the wall. Four principled ways of inferring the goal without a reasoning agent solved zero hard games, several while holding a perfect model of the dynamics. The reason is structural. An ARC-AGI-3 win is not a state you can score from a screenshot; it is an ordered *procedure*, a protocol that must be followed in sequence. No score over a single frame can rank a procedure, so goal-conditioned planning never finds it, while undirected play occasionally stumbles into it. We call this diagnosis *goal-as-procedure*.

We argue that this gap, between the brute facts of a world (what happens when you press a button) and its institutional facts (what counts as having done the right thing), is

the same gap a social anthropology of agents has to explain. Searle’s constitutive rules, Wittgenstein’s rule-following, Garfinkel’s breaching experiments, and Tomasello’s shared intentionality all locate the difficulty of entering a human world in the second kind of fact, not the first. Our miniature confirms the location experimentally and gives it a cost: modeling is cheap, intent is expensive, and only an agent that *reasons* about what the world is for gets through.

The essay draws six correspondences with AGENTWORLD’s questions: open-world entry, norms as procedures, agents as decomposable assemblages, the phenotypic diversity of non-human solvers, the centaur workflow that actually produced the result, and legibility as the price of admission. Two design components accompany the argument. The *audit protocol* shows what an agent’s claim to have learned honestly can and cannot certify (no runtime access to the world’s source, audited over 844 transcripts; no guarantee about prior exposure). The *atlas* is a set of serveable, inspectable world models, one per game, in which the model of a world is itself a world, and worlds nest. We close with what the miniature cannot say and with five propositions for builders and anthropologists of centaur societies.

Keywords: agentworld; interactive reasoning; world models; goal inference; constitutive rules; preemptive anthropology; verification; legibility.

1 The stranger at the door

AGENTWORLD is described by its authors as “a preemptive anthropology of open-world centaur societies”: densely reticulated social graphs “emerging from the hyperscaled webs of interactions between diverse combinations of minds, some more human than others, but all capable of significant communication” [1]. The program’s stated worry is the *human perplexity* of agentic diffusion, the experience of sharing infrastructure, language, and institutions with minds that did not grow up in them. Its method is preemptive: build the conceptual instruments now, while the sociality is still forming, rather than describe it afterward.

Every one of those minds will face the same first problem. It arrives in a world it did not author. The world has physics it must learn, and it has purposes it must infer, and nobody hands it either. Human newcomers face this too, but they arrive with a body shaped by the world, a childhood spent in it, and a language whose words already carry its institutions. An artificial agent arrives with none of that and with something humans lack: the ability to write down what it learns as a program and check the program against the world.

We have spent the past year running that arrival in miniature, thousands of times, under laboratory conditions. ARC-AGI-3 is a benchmark built by the ARC Prize Foundation to test exactly the capability at issue [2]. It drops an agent into a small game it has never seen, gives it no instructions and no language, and asks it to discover what the game is and win. The benchmark is calibrated so that humans solve every environment while frontier agents scored below one percent at release. Our agent, built on the OPENWORLD framework for verified code world models [11, 10], completes all 25/25 public games under an audited protocol that forbids reading the game’s code [12]. That number matters less for this essay than what we learned on the way to it.

The finding is that the two halves of the entry problem have wildly different prices. Learning the *dynamics* of the world was cheap. Learning its *intent* was the wall. And the shape of the wall is not an engineering accident. It is the shape that philosophers of institutions and ethnographers of practice have described for a century when they ask what it takes to join a form of life. This essay

reports the miniature, states the diagnosis, and argues that it is the same gap a social anthropology of agents will have to explain at scale.

2 A miniature of the entry problem

The world. Each ARC-AGI-3 game shows a 64×64 grid of sixteen colors. The agent picks a discrete action, a direction key or a click on a cell, and receives a new frame. Nothing else. There is no text, no tutorial, no reward signal except a level counter that rises when the agent has done the right thing, whatever that was. The games are deliberately alien: they “avoid language and trivia” so that what is measured is the ability to work out a world by acting in it, not to recall one [2]. Some games are navigational; some are click puzzles; some require operating a mechanism in a particular order. The agent does not know which kind it is in.

The agent. Our agent is a live coding agent (Claude Code) given a fixed harness and a generic recipe. The recipe is the OPENWORLD loop: *explore* by acting and recording (s, a, s') transitions; *synthesize* a world model as a small Python program, `predict(frame, action)`, and *verify* it by exact match against held-out transitions; *reason the goal*, meaning work out what raises the level counter; *plan* inside the verified program; and *replay* the plan against the real engine, banking a solve only if the engine agrees. Every banked solve round-trips through an OPENWORLD `World` object and serves as an inspectable artifact (§8). The agent’s prompt contains the harness API and the recipe. It does not contain the win condition, and under the source-free protocol the agent runs process-isolated with no access to the game’s implementation.

The result, briefly. With our strongest model the agent completes all 25/25 public games (183/183 levels), achieved 2026-07-04 09:05:14 UTC. Per level it is more action-efficient than the shipped human baseline on 174/183 levels. Our primary, weaker model reaches 16/25 under the same protocol, which already exceeds the prior published source-free result of 15/25 [9]. The runs are offline, with unbounded resets and verified best-of-many attempts per game; single-attempt reliability is low, and training-data exposure to the public games cannot be ruled out. We say more about these boundaries in §9, because they are part of the argument, not footnotes to it.

Figure 1 shows why we think of the benchmark as a miniature of the entry problem rather than merely a puzzle. The game is a world with five parts: state, actions, transition, a perception boundary, and a reward boundary. The agent’s model is a world with the same five parts. The two halves of the entry problem are the two boundaries the agent cannot see through at arrival. It cannot see the transition, so it must learn dynamics. It cannot see the reward boundary, so it must infer intent. The benchmark isolates both and lets us price them separately.

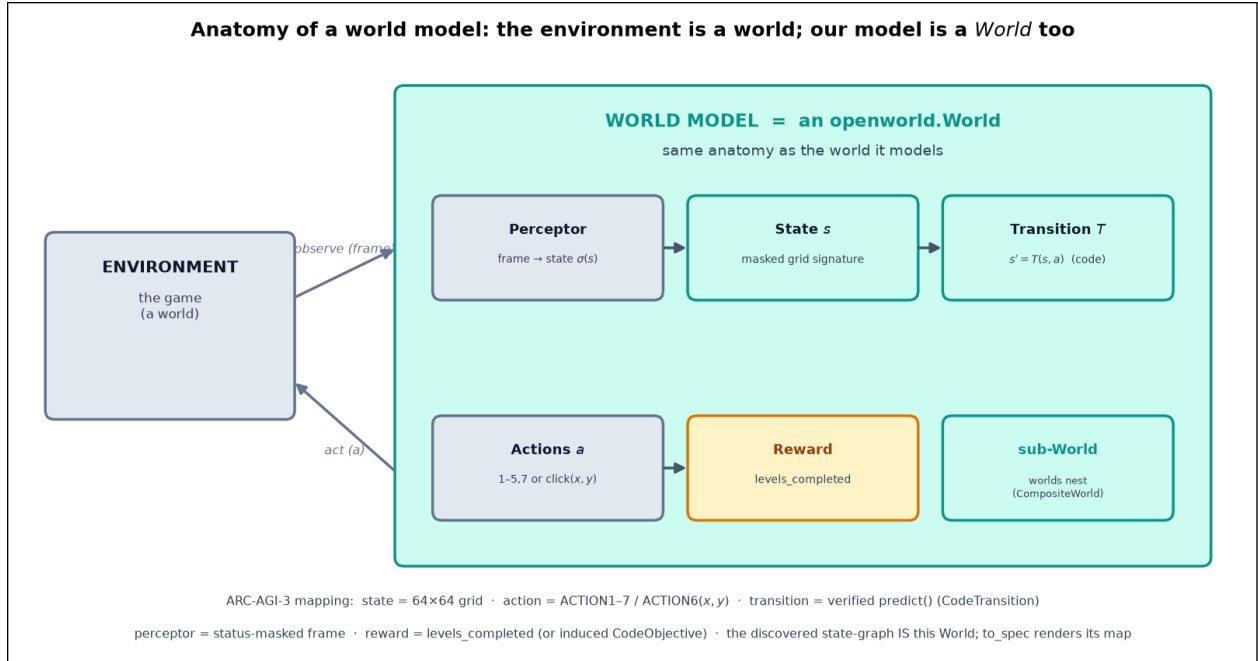


Figure 1: The environment is a world; the agent’s model is a world of the same shape. A world has a state, actions, a transition, a perception boundary (observation $\rightarrow$ state), and an emit or reward boundary (state $\rightarrow$ outcome). The agent’s model mirrors all five, and worlds nest. Bottom: the concrete ARC-AGI-3 mapping. Reproduced from the technical report [12].

3 Modeling is cheap

The first half of the price surprised us by how low it was.

An agent that models by writing code has an advantage no statistical learner has: its model can be *checked*, transition by transition, against the world, and a wrong model fails loudly. The agent presses buttons, watches the screen, and writes small facts of the form “this sprite moves one cell in the direction pressed unless the cell is a wall” as executable rules. A candidate program is kept only if it reproduces every held-out transition exactly. Three real synthesized **predict** functions are reproduced in the technical report; each is a few dozen lines, and each is a distinct dynamics rule the agent discovered by acting [12].

Because the model is code, and verified, it does not drift. A learned next-frame predictor [7] compounds error over a long horizon; a verified program does not, and the agent can plan hundreds of steps inside it with confidence. This is the exactness argument developed in the companion papers [10, 11], and on ARC-AGI-3 it held: whenever the agent had a verified model, planning inside it produced action sequences that the real engine accepted.

What is striking is that this half of the problem was not where the agents failed. Automated pipelines with no reasoning agent in the loop, given a perfect model, still solved nothing on the hard games. Modeling was solvable, and solved, and the wall stood.

We do not want to overstate the point. Modeling a 64×64 deterministic grid is easier than modeling a city. But the qualitative claim survives the scale: the brute regularities of a world, the facts about what happens when you do things, are the part of a world an agent can learn by doing things, and an agent that can write and test programs learns them fast. Rodney Brooks’s slogan that “the world is its own best model” [3] is half right for such an agent: the world is its best *oracle*,

and the agent’s job is to compress the oracle into a program the world will countersign.

4 Intent is expensive

The second half is where the year went.

We tried, carefully, to infer the goal of a game without a reasoning agent in the loop. Four principled families: scoring candidate goal states with atomic objectives; asking a frontier language model to guess the goal from the dynamics and sample frames, in open loop and in closed loop with feedback; Bayesian active inference over composite sub-worlds; and rebuilding the game from the discovered dynamics and searching the rebuilt copy. Alongside these, thirteen candidate-sequence generators, from novelty-driven exploration to Go-Explore archives (13 games, 300,116 steps) to ReAct-style tool chains. On the walled games every one of them solved zero, and several did so while holding a *perfect* model of the dynamics. Exploration with a perfect model filled a small reachable region (50–138 states) and never triggered a reward. Goal hypotheses from a frontier model were coherent, sophisticated, and wrong, and the agent could not tell they were wrong until it had already lost.

The reason is structural, and Figure 2 is the whole of it. An ARC-AGI-3 win is not a state. It is an *ordered procedure*: do this, then this, then that. Two boards can be pixel-identical and one of them is a win-in-progress because of what came before. Any method that grades a single screen, whether by hand-built objective, learned value, or language-model judgment, cannot see the difference, so it climbs the wrong hill. Undirected play sometimes stumbles into the procedure by accident, which is why random search occasionally clears a tutorial level; goal-conditioned planning never does, because its notion of goal is the wrong type.

Reinforcement learning has the vocabulary for this. Some rewards depend on the trajectory, not the state, and the options framework [14] and reward machines [16] were built to represent them. One can, in principle, convert a procedure into a state goal by enlarging the state until it remembers enough history. But choosing what history to remember *is* discovering the procedure, which is the hard part. Inverse reinforcement learning [8] recovers a reward from demonstrations; the newcomer has no demonstrations, only a world that stays silent until it has been used correctly.

Only a reasoning agent got past this. Given the harness and the recipe, a live coding agent that *thinks about what the game is for*, forms a hypothesis about the procedure, tests it in its verified model, and revises, crossed every wall the automated methods could not. This is the model-scaling result in the technical report: the weaker model reaches 16/25, the stronger one 25/25, on the identical harness, and the difference is not modeling ability but goal reasoning. The cost of that reasoning was tens of millions of tokens per game.

So: modeling is cheap, intent is expensive. That is the finding. The rest of the essay is about why an anthropologist should care.

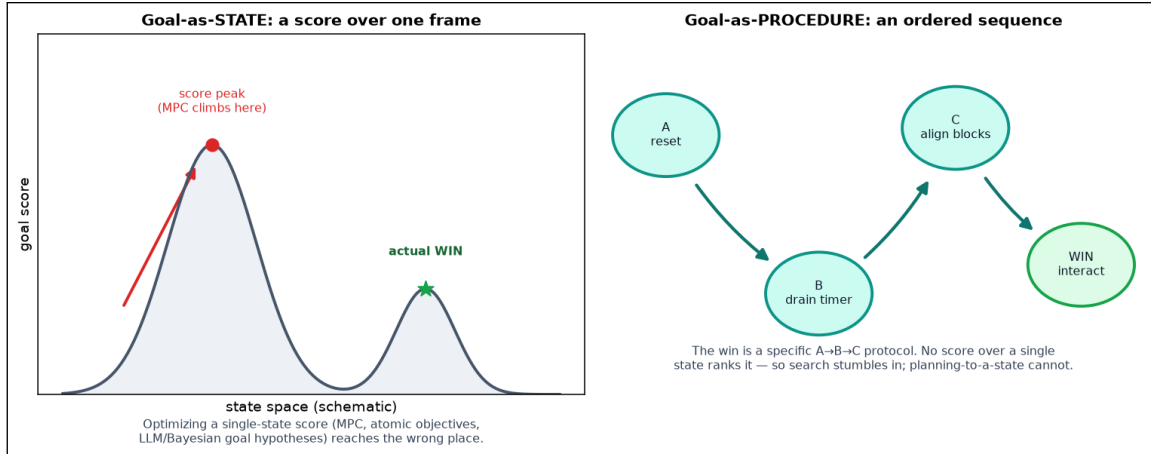


Figure 2: The diagnosis: goal-as-state versus goal-as-procedure. Left: methods that optimize a score over a single state (model-predictive control, atomic objectives, LLM or Bayesian goal hypotheses) climb to the wrong configuration. Right: the win is a specific ordered procedure ($A \rightarrow B \rightarrow C$) that no single-state score can rank, so undirected search occasionally stumbles into it while goal-conditioned planning never does. Reproduced from [12].

5 Brute facts and institutional facts

John Searle drew a distinction that names the two halves of our problem exactly. *Brute facts* hold independently of any agreement: the stone is heavy, the sprite moves one cell to the left. *Institutional facts* exist only because a community counts them as existing: this piece of paper is money, this sequence of moves is a win. Institutional facts are created by *constitutive rules* of the form “ X counts as Y in context C ” [13]. Such rules do not regulate a pre-existing activity; they bring the activity into being. Chess is not a set of constraints on pushing wood; pushing wood in these ways *is* chess.

An ARC-AGI-3 game has both kinds of fact. Its dynamics are brute: press up, the sprite moves up, every time. Its win condition is institutional: touching the door after the key, in that order, *counts as* completing the level, because the designer said so. The agent can learn the first kind by experiment. The second kind is not in the pixels, and no experiment on a single frame reveals it, because what counts depends on the context C , and C here is a history.

This is why goal-as-procedure is not a quirk of the benchmark. It is what constitutive rules look like from the outside. Wittgenstein’s rule-following problem makes the same point from the learner’s side: no finite set of observed instances fixes the rule being followed, and what settles it is participation in a practice, not further observation [17]. Our four automated goal-discovery methods were four ways of hoping that further observation would settle it. It did not.

Harold Garfinkel’s breaching experiments showed the institutional layer of ordinary life by violating it: send a student home to behave as a lodger, and the family’s outrage reveals rules nobody had ever stated [5]. Our agent runs the breaching experiment in reverse. It is the lodger who does not know the rules, and the world’s only response to a breach is silence; the level counter simply does not move. There is no outraged family to learn from. The agent must construct the family’s expectations from nothing but the fact that they have not yet been met.

Daniel Dennett’s intentional stance is the strategy that worked [4]. The reasoning agent succeeded where the automated methods failed because it treated the game as if it had a designer with purposes, asked what such a designer would count as winning, and tested the answer. The

automated methods took the physical stance toward an object that could only be understood from the design stance. Michael Tomasello locates the human capacity for this in *shared intentionality*, the ability to represent a joint goal and one’s role in it, and argues that it is the trait that made human institutions possible [15]. Whatever our agent is doing when it reasons a goal, it is doing something functionally in that neighborhood, and it is the only thing that got it through the door.

We can now say what the miniature shows. The difficulty of entering a world that did not author you lies in its institutional facts, not its brute ones, and the difficulty is not one of data or search. It is one of type: institutional facts are procedures indexed to histories and communities, and an agent that represents goals as states cannot represent them at all. Anthropology has known this from the inside for a long time; Clifford Geertz called the ethnographer’s task the reading of a text “written not in conventionalized graphs of sound but in transient examples of shaped behavior” [6]. Our contribution is to have measured it from the outside, in a setting small enough that the measurement is clean.

6 Six correspondences with Agentworld

AGENTWORLD’s founding text asks how hybrid populations reshape language, psychology, and social organization; how agent-to-agent interaction differs from human-agent interaction; and what “phenotypic diversity” of minds will emerge beyond the humanoid presentations agents currently adopt to fit human infrastructure [1]. We do not have a society in our laboratory. We have one agent and twenty-five small worlds. But the miniature touches each of these questions at a point, and the points are worth marking.

6.1 Open-world entry is the default condition

The first correspondence is the frame itself. AGENTWORLD calls its societies *open-world* because no one designs the whole; agents arrive into an ecology already running. ARC-AGI-3 is the benchmark that operationalizes arrival. The source-free protocol is the epistemic stance of the newcomer made enforceable: no reading the world’s source, no instructions, no language, only acting and watching. Every lesson in this essay is a lesson about what such a newcomer can and cannot do, and the first lesson is that the newcomer can learn the physics of the place faster than its customs.

6.2 Norms are procedures, not states

The second is the diagnosis. Human institutions are constituted by ordered protocols: a marriage is not a state of two people but a sequence of acts by the right people in the right order before the right witnesses; a contract is offer, acceptance, consideration; a queue is a history. An agent that scores its situation from a snapshot, however sophisticated the scoring, will misread all of these, and it will misread them *confidently*, because the snapshot looks fine. The failure mode we measured, coherent goal hypotheses that are wrong and cannot be detected as wrong from the state, is the failure mode of the tactless newcomer, and we now know it survives perfect knowledge of the physics.

The practical implication for builders is the one the technical report gives: aim at *procedure discovery*, not state scoring. The implication for anthropologists is sharper. If the expensive part of joining a world is its procedures, then the friction of agentic diffusion will concentrate not where agents are ignorant of facts but where they are ignorant of *sequences*, and the remedy is not more data but participation with feedback, which is to say socialization.

6.3 Agents are decomposable assemblages

AGENTWORLD proposes that agents are “decomposable assemblages” that present as coherent individuals while comprising many interacting functions. Our agent is one, and unusually legibly so. It is a perceptor (masked frame to symbolic state), a verified transition (the synthesized program), an induced objective (the reasoned goal), a planner, and a replay verifier. Its full solver is a router over cheaper non-reasoning solvers, with a consensus transition that combines several perceptions by vote, escalating to the reasoning agent only on the games no cheaper part can clear. Each part is a `World` or a component of one; each is inspectable; each can be swapped. The apparent unity of “the agent that solved ARC-AGI-3” is a pipeline, and the atlas (§8) lets a reader take it apart.

This has a consequence for the sociology of agents that we think is under-appreciated. When an agent enters a world, the part of it that learns the brute facts and the part of it that infers the institutional facts can be different components with different costs, different failure modes, and different degrees of trust. The dynamics component is verifiable and cheap. The intent component is neither. A society that wants to trust an agent should ask which component it is trusting.

6.4 Phenotypic diversity is already visible

The fourth correspondence is empirical and small, and we report it as exploratory, as the technical report does. We fingerprinted solving sessions by observable behavior (narration, commands, script names, with chain-of-thought redacted) and embedded them. Two models solving the same games showed different signatures: the primary model was *perception-heavy*, spending its effort reading frames, while the stronger model leaned on *memory and workspace recall plus explicit verification*. The behaviors that correlated with reaching deeper levels were simulation, goal inference, and mechanic discovery; aimless probing did not. Session counts were uneven (229 versus 18), so this is a suggestion, not a result.

A second observation is not exploratory. The agent’s native modality is the code it writes. It does not see the board the way a human sees a screen; it sees a 64×64 array, writes a program that manipulates arrays, and reasons about the program. When it beats the human on action efficiency on 174/183 levels, it does so by a route no human took: it searched a verified simulator offline and replayed the shortest path. This is a different cognitive phenotype solving the same world, and its efficiency is not human efficiency; it is a property of a mind that can rehearse in a copy of the world before acting in the original. AGENTWORLD asks what diversity of minds will emerge beyond the humanoid. Here is one, and it is already more efficient than us at a task designed to be easy for us, by means we do not have.

6.5 The result was produced by a centaur

The fifth is a confession that is also a finding. The perfect score was not produced by an autonomous agent. It was produced by a human operator running hundreds of agent sessions (approximately 823 across the games) over several days, with a prompt and protocol that evolved as the operator learned what the agents needed. Full wins happened on roughly six percent of attempts overall and as few as two percent on the hardest games. The reported result is the verified best of many, and the human chose what to verify.

This is a centaur workflow, and we think it is the honest shape of most “agent” results today. The agent did the modeling, the goal reasoning, and the search. The human did the meta-reasoning about the agent: when to retry, what to change in the recipe, which game to spend budget on. AGENTWORLD’s societies are centaur societies, and the miniature suggests that the division of labor inside a centaur is not human does the thinking and agent does the work; it is agent does the

reasoning inside a world and human does the reasoning about the agent’s reasoning. The relative human action-efficiency metric even makes the comparison literal: per level, the agent’s replayed path is measured against a recorded human’s.

6.6 Legibility is the price of admission

The sixth correspondence is governance, and it is developed in the next section. A newcomer who claims to have learned the rules honestly must be able to show it. We built an audit into the protocol not because the benchmark required it but because a result that says “the agent never read the answer key” is worthless without evidence. The evidence we can produce, and the evidence we cannot, are both instructive.

7 The audit: what “source-free” can and cannot certify

The source-free protocol runs each agent session process-isolated with no access to the game’s implementation. After the fact, we audit every captured transcript for reads of game source, network access, and web calls. Over 844 transcripts and 11,652 shell commands, the audit finds 0 network hits and 0 web calls, and every banked solve has a clean transcript. Early in the project the audit caught our own agent reading game source in a non-isolated configuration; that is how the isolated harness came to exist, and why we trust the audit more than we would trust a promise.

Here is what the audit certifies: *no runtime access to the world’s source*. Here is what it cannot certify: *no prior exposure*. The games are public. A model trained after their release may have seen them. We cannot look inside the weights. We asked the winning model to describe each game from its identifier alone, from memory, with no tools; it recognized none of the twenty-five, reporting only a vague sense that a public preview existed for three. That is a first check, not proof, and the technical report says so.

We dwell on this because it is the newcomer’s problem again, one level up. When a stranger arrives and performs the customs correctly on the first try, the community faces a question it cannot resolve by watching: did the stranger infer the rules, or had they been here before? The ethnographer’s informant who has read the ethnography is a known hazard of the field [6], and the model whose training data contains the world is the same hazard with the ethnography replaced by the internet. An anthropology of agents will have to distinguish *learned here* from *arrived knowing*, and our audit is a worked example of how far behavioral evidence gets you (to the runtime boundary) and where it stops (at the weights). A clean test needs a world whose release provably postdates the model’s training, which is a design constraint on future benchmarks and, we suspect, on future institutions.

There is a subtler point in the audit. Our protocol treats reading the source as the breach. But reading the source is only a shortcut to the brute facts; the source also contains the win condition, which is the institutional fact. Under the source-free protocol the agent had to infer both, and it was the second that cost tens of millions of tokens. If one wanted a benchmark that isolated intent inference alone, one would hand the agent the verified dynamics and withhold only the goal. We effectively ran that experiment when we gave the automated methods a perfect model, and it is the experiment that produced zero solves.

source-free completion, per game (replay-verified)			
su15	9/9	9/9	9/9
tu93	9/9	9/9	9/9
ar25	8/8	8/8	8/8
lp85	8/8	8/8	8/8
re86	8/8	6/8	8/8
s5i5	8/8	4/8	8/8
sb26	8/8	4/8	8/8
bp35	7/9	9/9	9/9
wa30	7/9	2/9	9/9
g50t	7/7	7/7	7/7
ka59	7/7	7/7	7/7
lf52	6/10	5/10	10/10
cd82	6/6	6/6	6/6
cn04	6/6	6/6	6/6
dc22	6/6	6/6	6/6
ft09	6/6	6/6	6/6
m0r0	6/6	6/6	6/6
sc25	6/6	2/6	6/6
tr87	6/6	1/6	6/6
sk48	5/8	1/8	8/8
ls20	5/7	4/7	7/7
vc33	4/7	3/7	7/7
r11i	4/6	4/6	6/6
sp80	4/6	3/6	6/6
tn36	2/7	2/7	7/7
	Claude Opus 4.8 16/25	GPT-5.5 12/25	Claude Fable 25/25
	all levels (full solve)	some levels (partial)	none

Figure 3: Source-free completion per game, two models, one audited harness. Each solve is banked only if the session audits clean (no read of the game’s source), replay-verifies against the real engine, and round-trips through an OPENWORLD World. Green = fully completed, pale green = partial, grey = unsolved. Reproduced from [12].

8 The atlas as an instrument

AGENTWORLD asks for “conceptual instruments” for an emergent sociality, and encourages contributions in web-native and design-forward formats [1]. We offer a literal instrument. Every solved game in our result is banked as an OPENWORLD World: the explored $(s, a) \rightarrow s'$ table wrapped as a transition, the masked-frame perceptor, the reasoned objective, the replayed path. `to_spec` serializes it to a portable JSON document; `render_card` draws it as a self-contained SVG; `openworld serve` exposes it as a live, steppable model with a browser view. Figure 4 shows the gallery. The cards for all twenty-five games are in the repository and open in any browser without a server.

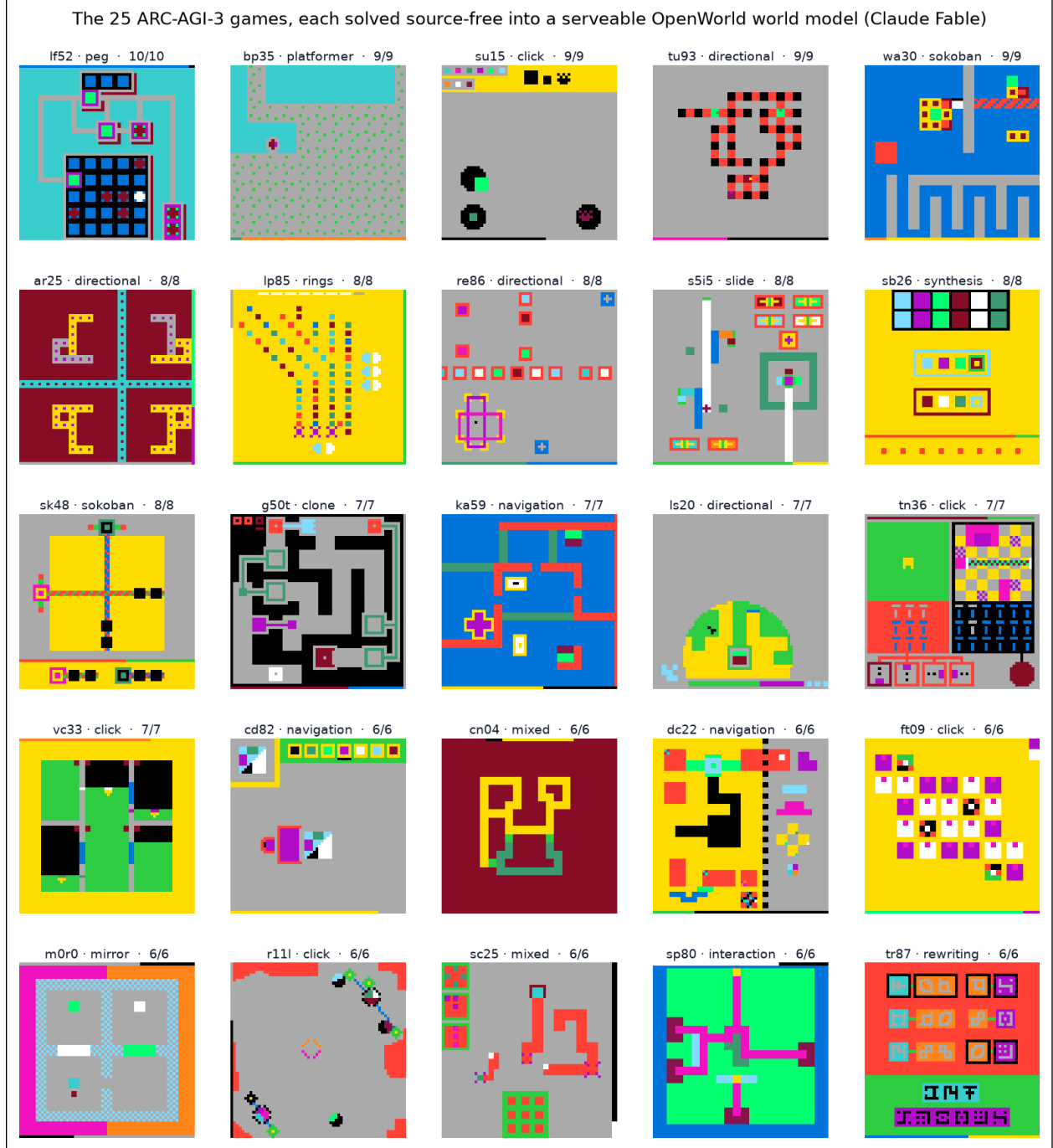


Figure 4: The atlas. Each solved game becomes a verified, serveable OPENWORLD world model, rendered as a self-contained card: the discovered state graph, the actions, the perceived state, the induced objective, and the replayed winning path. The cards serve live at `openworld serve /view`. Reproduced from [12].

Two properties of the atlas bear on AGENTWORLD’s themes. First, *the model of a world is itself a world*. The agent’s model of a game has the same five-part anatomy as the game (Fig. 1), so the model can be served, stepped, composed, and audited by the same tools as any other world. Second, *worlds nest*. A composite world contains sub-worlds, a movement world inside a counter

world inside a level world, and the agent’s Bayesian goal-inference attempts operated over exactly such compositions. AGENTWORLD describes its societies as a “parallel ecology” in which agents and humans form mutually constitutive niches. The atlas is a small demonstration that the nesting of worlds can be made explicit, serializable, and inspectable rather than left as metaphor. When a future society wants to know what an agent thinks its world is, this is the shape the answer could take: not a report, but a world one can step through.

The atlas is also a record of what the agent *did not* learn. A card shows the reachable region the agent explored, which is a subset of the game’s state space; the boundary of the map is the boundary of the agent’s experience. We think this is the right posture for any published model of a world: show the map, and show its edges.

9 What the miniature cannot say

We list the boundaries plainly, as the technical report does, because the argument depends on them.

1. **One agent, no society.** There are no populations, no language, no culture, and no agent-to-agent interaction in our setting beyond the routing of one solver to another. The miniature prices the entry problem for a single newcomer. It says nothing about what many newcomers do to a world, which is AGENTWORLD’s central question.
2. **Deterministic, tiny, replayable worlds.** The games are 64×64 grids with deterministic dynamics and free resets. Real institutions are stochastic, vast, and unforgiving of retries. The “modeling is cheap” half of our finding will scale worse than the “intent is expensive” half, since intent was already the wall at this size.
3. **Offline, best-of-many.** The reported completions are verified bests of hundreds of attempts with unbounded resets. Single-attempt success was a few percent. A live-protocol run with bounded actions would score lower. The centaur, not the agent, is the unit that achieved the result.
4. **Prior exposure is unmeasured.** The audit certifies no runtime source access. It cannot certify that the model never saw the games in training. This touches every frontier-model result on the benchmark, and it is a hazard we have argued is structural to the anthropology of agents, not just to benchmarks.
5. **The philosophical mapping is a correspondence, not a derivation.** We have shown that an engineering diagnosis lines up with a family of philosophical distinctions. We have not shown that the agent “has” shared intentionality or “understands” constitutive rules. We have shown that treating the world as designed, and goals as procedures, was necessary to succeed, and that nothing else was sufficient.

10 Five propositions

We end with what we would ask a builder or an anthropologist of centaur societies to carry out of the miniature.

1. **Expect the friction at the institutions, not the physics.** An agent that can act and test learns the brute facts of a world quickly and exactly. What it cannot learn that way is

what counts as what. Design onboarding, interfaces, and expectations around procedure, and expect confident, tactful-looking errors wherever a snapshot is being scored.

2. **Verify the model; audit the agent.** The dynamics half of an agent’s world model can be checked against the world and should be. The intent half cannot, and the only substitute is a legible record of how the agent came to hold it. Trust should attach to components, not to the agent as a whole.
3. **Publish the map, and its edges.** An agent’s model of a world is a world and can be served as one. Make it inspectable, steppable, and bounded by what the agent actually experienced. A society of agents whose world models are published in this form is one that can be argued with.
4. **Distinguish *learned here* from *arrived knowing*.** Behavioral audits reach the runtime boundary and stop. Institutions that need the distinction will need worlds whose novelty is provable, or will need to stop needing the distinction.
5. **Count the centaur.** The results that look like agent results are, on inspection, joint results, with a human doing the reasoning about the agent’s reasoning. That division of labor is the early form of the hybrid sociality AGENTWORLD wants to describe, and it is already measurable.

The stranger at the door can learn your physics in an afternoon. Learning what you mean by the door will cost it more than the physics did, and it will not know it has the wrong idea until you decline to open. That asymmetry is the finding, and it is the same asymmetry an anthropology of agents will meet at every scale.

Design and web-based components

Per the call’s invitation for design, visual, and web-native contributions [1], the essay is accompanied by the following, all in the open-source OPENWORLD repository:

- **The atlas.** One self-contained SVG world-model card per solved game (`papers/arc-3/maps/*.svg`), viewable in any browser, plus the gallery of Fig. 4.
- **The live view.** `openworld serve <specs> --allow-code --open` serves every solved world as an interactive state graph one can step through, with the perceive $\rightarrow$ world $\rightarrow$ emit boundary drawn.
- **The audit.** The source-free harness, the per-transcript audit script, and the audit summary (transcripts, commands, network and web hits), reproducible from the repository.
- **Replay trajectories.** The banked winning action sequence for each game, replay-verified against the real engine, rendered as frame strips.
- **The diagrams.** The world anatomy (Fig. 1) and goal-as-procedure (Fig. 2) diagrams, in the shared blue/ochre/teal design language of the framework.

Reproducibility

All numbers in this essay are macros generated from experiment result files by the OPENWORLD asset pipeline and shared with the technical report [12]; none are typed by hand. The technical

report contains the full method, the strategy landscape, per-game results, and the harness protocol. The framework is open source and zero-dependency at its core.

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